

ORIGINAL CONTRIBUTION

Preoperative Predictive Modeling of Recurrent Graft Failure: Development and Validation of a 12-Month Prognostic Tool in Kidney Transplant Recipients

Tongtong Shen^a, Xinyao Wang^{b*}, Xueyu Liu^a, and Wen Zheng^{b*}

^aCollege of Artificial Intelligence, Taiyuan University of Technology, Jinzhong, Shanxi, China; and ^bUniversity of Chinese Academy of Sciences, Beijing, China

ABSTRACT

Background. Kidney transplantation is among the most effective treatments for end-stage renal disease. However, kidney transplant (KT) recipients remain at a relatively high risk of mortality, with underlying causes yet to be fully elucidated. This study aim to develop a machine learning model to predict 12-month graft survival in KT recipients and to analyze the relationship between key postoperative indicators and graft survival.

Methods. We retrospectively collected clinical data from 368 KT recipients who underwent allogeneic transplantation between April 2016 and September 2022. A novel feature selection method was employed to exclude variables with weak correlations to outcomes. Eight machine learning algorithms were compared to develop an optimal predictive model, with a primary focus on efficiency.

Results. Among the evaluated models, the light gradient boosting machine model demonstrated superior predictive performance, achieving an area under the receiver operating characteristic curve of 0.80 and 0.71 for internal and external validations, respectively. Notably, we redefined the threshold for assessing renal dysfunction in KT recipients, proposing an estimated glomerular filtration rate of 45 mL/min/1.73 m² as a more precise standard.

Conclusion. The proposed machine learning model and the newly established estimated glomerular filtration rate threshold provide accurate assessments of graft dysfunction severity, offering robust support for clinical decision-making in kidney transplantation management.

CHRONIC kidney disease (CKD) is a progressive condition characterized by a gradual decline in renal function, ultimately leading to a cascade of complications and multiorgan failure [1,2]. As kidney function deteriorates further, CKD progresses to end-stage renal disease (ESRD), defined clinically by a glomerular filtration rate (GFR) below 15 mL/min/1.73 m², at which point the kidneys can no longer clear metabolic waste effectively. While early CKD can be managed to restore renal function, ESRD represents an irreversible state that necessitates renal replacement therapy. Over the past two decades, kidney transplantation, hemodialysis, and peritoneal dialysis have become established therapeutic options for ESRD management [3], with kidney transplantation emerging as the preferred approach due to its convenience and long-term benefits [4,5].

According to reliable data, the limited availability of donor organs remains a significant barrier to transplantation, with current demand estimates of 2692 cases per million population in Japan (2015) [6], 1700 cases per million in the United States (2009) [7], and 782 cases per million in the European Union (2013) [8].

For ESRD patients, kidney transplantation does not entirely resolve the issue of renal dysfunction. The vulnerability of grafts and the compatibility challenges between donors and

*Address correspondence to Xinyao Wang and Wen Zheng, University of Chinese Academy of Sciences, 19(A) Yuquan Road, Shijingshan District, Beijing 100049, China. E-mails: wangxinyao@microsat.ac.cn, zhengwen@microsat.ac.cn

recipients often result in early graft failure, placing patients at risk of subsequent graft loss. This not only fails to restore renal function effectively but also represents a potential waste of scarce donor resources [9,10]. Studies have shown that kidney transplant (KT) recipients exhibit significant clinical differences at various postoperative time points, closely linked to graft survival outcomes. These observations have spurred substantial interest in unraveling the mechanisms underlying graft survival.

Traditional medical analyses, particularly in nephrology, have primarily relied on experimental design and statistical theories. However, with the advent of machine learning, researchers have increasingly applied computational methods to the field of kidney transplantation. For instance, Mark et al [11] developed a survival prediction model using statistical techniques, successfully identifying key prognostic factors and achieving superior performance compared to the survival models used in the US Kidney Allocation System. Similarly, Decruyenaere et al [12] applied various machine learning algorithms to predict delayed graft function, finding that linear support vector machines (SVMs) achieved the highest predictive accuracy. Ravindhran compared machine learning algorithms with traditional regression methods, demonstrating that machine learning could handle complex nonlinear data and significantly improve prognostic accuracy [13]. These studies primarily employed classical machine learning models, which are well-regarded for their strong mathematical foundations, reliability, and stability.

In the field of graft survival prediction, numerous survival analysis studies have explored a wide range of variables, ranging from six to several hundred [14–17]. For instance, Lin et al [18] employed regression models and artificial neural networks (ANN) to predict graft survival, but their work lacked a focus on model interpretability. While advanced machine learning approaches have achieved high predictive accuracy in graft survival studies, many fail to thoroughly analyze relevant biomarkers, leaving the intricate mechanisms of graft survival largely unexplained [19–21].

Globally, GFR is a widely accepted metric for evaluating renal function, serving as a measure of kidney filtration capacity and overall health. However, due to the complexity and invasive nature of GFR measurement, many researchers rely on creatinine-based estimations, such as the estimated GFR (eGFR), as a surrogate. The eGFR threshold of 60 mL/min/1.73 m² is commonly used to define renal dysfunction in the general population and has been widely adopted in clinical practice [22–24]. However, this threshold is unsuitable for KT patients, whose renal function is inherently more fragile than that of healthy individuals. Systematic studies to establish a tailored standard for evaluating renal dysfunction in KT recipients are lacking. This study seeks to address this gap by identifying an eGFR threshold specific to KT patients, enabling more accurate differentiation of renal dysfunction, early intervention, and improved graft function preservation.

In this study, we aimed to leverage machine learning to explore the relationships between preoperative characteristics, key postoperative indicators, and graft survival. Additionally, we identified an optimal eGFR threshold for assessing renal

dysfunction in KT recipients. Specifically, we developed a lightweight and efficient tree-based model to predict 12-month graft survival in KT patients.

MATERIALS AND METHODS

Patient Selection and Study Designs

This single-center retrospective study was conducted at a hospital in China with extensive experience in kidney transplantation and comprehensive clinical data management. Clinical records of patients evaluated for kidney disease between April 2016 and September 2022 were reviewed, yielding 513 KT recipients. All data were anonymized to ensure patient confidentiality. Trained personnel extracted standardized variables, including demographics, laboratory results, and other clinical parameters. Patients were followed up to assess post-transplant outcomes, focusing on 12-month graft survival. Eligible cases meeting the inclusion and exclusion criteria constituted the final dataset for model development (Fig 1).

Inclusion and Exclusion Criteria

The study population comprised patients who had undergone kidney transplantation for ESRD, defined as a failure of both kidneys to function. Patients who were still in the hospital for follow-up examinations were included in the study.

- (1) It is also important to consider the possibility of coinfection with other viruses, including hepatitis C virus, hepatitis D virus, hepatitis E virus, and human immunodeficiency virus.
- (2) Complicated with liver cancer or other malignant tumors.
- (3) Conjunction with a multitude of terminal illnesses and systemic inflammation, including persistent infection, respiratory failure, decompensated heart failure, and the onset of autoimmune diseases.
- (4) Insufficient clinical data in medical records.

Overview of Machine Learning Models

In this study, eight machine learning models were employed to fit real-world patient data, each briefly described as follows. Logistic regression is an algorithm that derives class probabilities using a logistic function, allowing for the evaluation of relationships between the target variable and multiple features [25]. SVM achieves effective separation of samples from different classes by identifying an optimal hyperplane. Naive Bayes is a probabilistic classifier based on Bayes' theorem, assuming feature independence and modeling the probability distribution of continuous features using a Gaussian distribution. ANN simulates biological neural networks, enabling the capture of complex nonlinear relationships among features. Random forests are ensemble learning models comprising multiple decision trees, where each tree independently makes predictions, and the final output is determined through aggregation. Extreme gradient boosting (XGBoost) constructs an efficient and powerful classifier by iteratively training multiple weak learners and

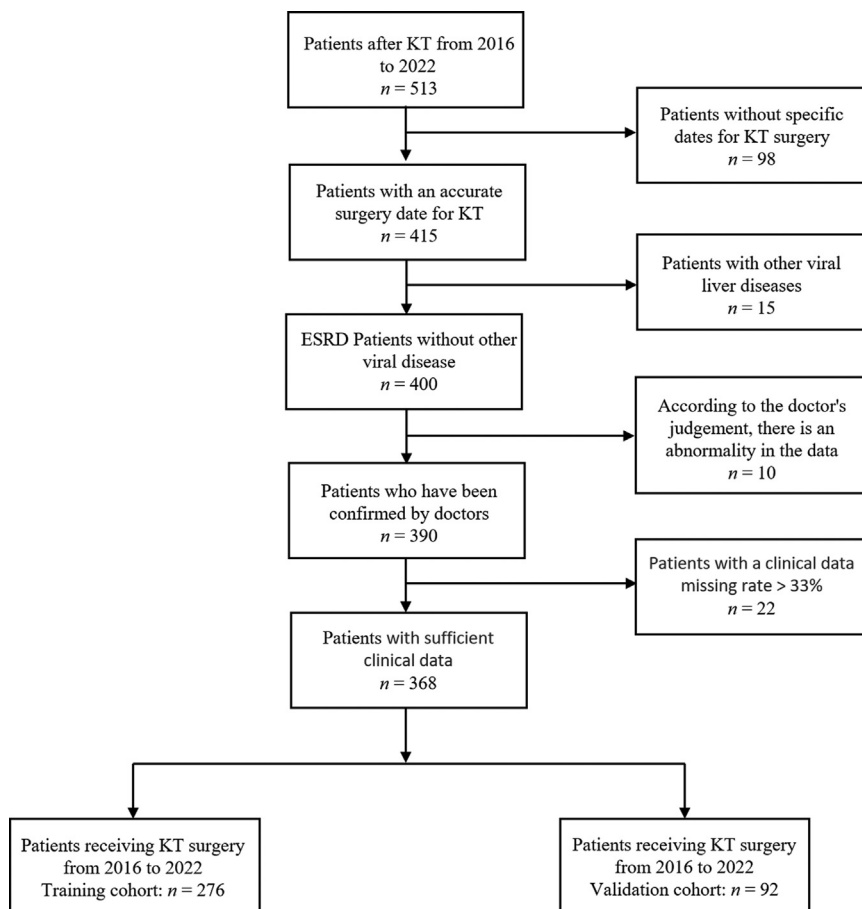


Fig 1. Flow diagram of the study population.

optimizing the loss function using gradient descent [26]. Light gradient boosting machine (LGBM) is a highly optimized gradient boosting algorithm that utilizes histogram-based learning to markedly improve both training efficiency and predictive accuracy. Similarly, CatBoost, another advanced gradient boosting framework, leverages a symmetric tree-growing technique to enhance computational performance and achieve robust model generalization. For all model training processes, exhaustive grid search and 10-fold stratified cross-validation were applied to determine the optimal hyperparameters for each model.

Statistical Analysis

Patient data were presented as continuous and categorical variables. The Kolmogorov–Smirnov test was used to assess whether continuous variables followed a normal distribution. For variables with a normal distribution, data were expressed as mean $\pm$ standard deviation, while non-normally distributed variables were presented as medians [27,28]. Categorical variables were reported as frequencies and percentages. Additionally, Spearman's rank correlation test was employed to evaluate the ordinal relationship between two variables. All statistical tests

were two-tailed, and a P -value of $<.05$ was considered statistically significant. Data analysis was conducted using the SPSS statistical software package (version 25.0; SPSS Inc).

RESULTS

Patients' Characteristics

A review of clinical data provided by a hospital identified 513 KT patients. Based on the predefined inclusion and exclusion criteria, a total of 356 eligible patients were selected. These patients were subsequently divided into a training cohort (267 cases) and an internal validation cohort (89 cases). The baseline characteristics of the cohorts are presented in Table 1.

In the training cohort, the median age was 38.0 years, with 70.4% of the patients being male. Among these patients, 3.0% had diabetes, and 9.0% had primary hypertension. In the internal validation cohort, the median age was 42.0 years, with 62.9% being male; 2.2% of patients had diabetes, and 9.0% had primary hypertension. Notably, 69 patients (25.8%) in the training cohort experienced graft failure within 12 months, while 20 patients (22.5%) in the internal validation cohort also experienced graft failure during the same period.

Table 1. Baseline Features of Training Set (*n* = 267) and Verification Set (*n* = 89)

Characteristics	Training Cohort (<i>n</i> = 267)	Validation Cohort (<i>n</i> = 89)
Male, <i>n</i> /total (%)	188/267 (70.4)	56/89 (62.9)
Age (y)	38.0 (31.0-49.0)	42.0 (33.0-50.5)
Age <40 y, <i>n</i> /total (%)	138/267 (51.7)	39/89 (43.8)
Age ≥40 y, <i>n</i> /total (%)	129/267 (48.3)	59/89 (66.3)
Blood type A	73/267 (27.3)	30/89 (33.7)
Blood type non-A	194/267 (72.7)	59/89 (66.3)
Kinship renal	146/267 (54.7)	49/89 (55.1)
Allogeneic living kidney	13/267 (4.9)	3/89 (3.4)
Dead body renal	108/267 (40.4)	37/89 (41.5)
Diabetes mellitus, <i>n</i> /total (%)	8/267 (3.0)	2/89 (2.2)
Hypertension, <i>n</i> /total (%)	24/267 (9.0)	8/89 (9.0)
HDL C (mmol/L), <i>n</i> /total (%)	1.39 (1.13-1.59)	1.40 (1.20-1.63)
HDL C ≥ 1.40 mmol/L, <i>n</i> /total (%)	137/267 (51.3)	44/89 (49.4)
HDL C < 1.40 mmol/L, <i>n</i> /total (%)	130/267 (48.7)	45/89 (50.6)
BUN (mmol/L), <i>n</i> /total (%)	8.10 (6.38-10.18)	7.42 (6.05-9.38)
BUN > 7.80 mmol/L, <i>n</i> /total (%)	141/267 (52.8)	39/89 (43.8)
BUN < 7.80 mmol/L, <i>n</i> /total (%)	126/267 (47.2)	50/89 (56.2)
HGB (g/L), <i>n</i> /total (%)	125.0 (112.0-141.0)	124.5 (116.8-141.0)
HGB ≥ 120 g/L, <i>n</i> /total (%)	172/267 (64.4)	57/89 (64.0)
HGB < 120 g/L, <i>n</i> /total (%)	95/267 (35.6)	32/89 (36.0)

Building High-Performance Predictive Models With Machine Learning

First, parameters with more than 35% missing values were excluded from the analysis, leaving a final selection of 15 variables with complete data. Second, traditional machine learning studies often identify the final set of features by iterative modeling, progressively eliminating less important variables in each iteration. In contrast to this conventional approach, we utilized Spearman's rank correlation test to evaluate the association between each feature and the outcome. Finally, applying the criteria of a correlation coefficient ≥ 0.1 and a *P*-value < 0.05 , we identified six most relevant features: sex, blood type, donor source, high-density lipoprotein cholesterol (HDL-C), blood urea nitrogen (BUN), and hemoglobin (HGB). These variables were used in the training cohort and applied to the internal validation cohort.

Using the six selected features, we constructed streamlined models capable of accurately predicting 12-month graft outcomes in KT patients. The best-performing model was identified based on evaluation metrics, including accuracy, precision, recall, F1-score,

and area under the curve (AUC), shown in Table 2. In the training cohort, the AUC values of the eight machine learning models ranged from 0.49 to 1.0, while in the internal validation cohort, the AUC values ranged from 0.42 to 0.74 (Fig 2). Specifically, SVM and ANN demonstrated relatively poor performance, likely due to their limited capacity to handle the complex nonlinear relationships inherent in medical datasets. Despite their simplicity, the LR and Naive Bayes models exhibited good performance when fitting our data. The remaining four models demonstrated higher predictive performance, with statistical tests indicating no significant differences in performance between LGBM and random forest, XGBoost, or CatBoost. Based on the combined results from the training and validation cohorts, the LGBM model was selected as the final predictive model.

Internal and External Validation of the Predictive Model

The LGBM model demonstrated a consistently positive net benefit across probability thresholds between 0.1 and 0.7 in both

Table 2. Diagnostic Performance of Different ML Models for Survival After Kidney Transplantation

Model	Accuracy	Precision	Recall	F1-Score	AUC
LogisticRegression	0.6676	0.6835	0.7743	0.7261	0.7226
SVM	0.3812	0.2998	0.2416	0.3504	0.3418
Naive Bayes	0.6976	0.7212	0.7588	0.7395	0.7197
ANN	0.6427	0.7878	0.6410	0.7069	0.6064
RandomForest	0.8277	0.7378	0.7715	0.7543	0.7804
XGboost	0.7697	0.7715	0.7214	0.7456	0.8014
LightGBM	0.7773	0.8078	0.8128	0.8103	0.8047
CatBoost	0.7377	0.7888	0.8002	0.7945	0.7992

Bold values indicate the best-performing values for each diagnostic performance metric among the compared models.

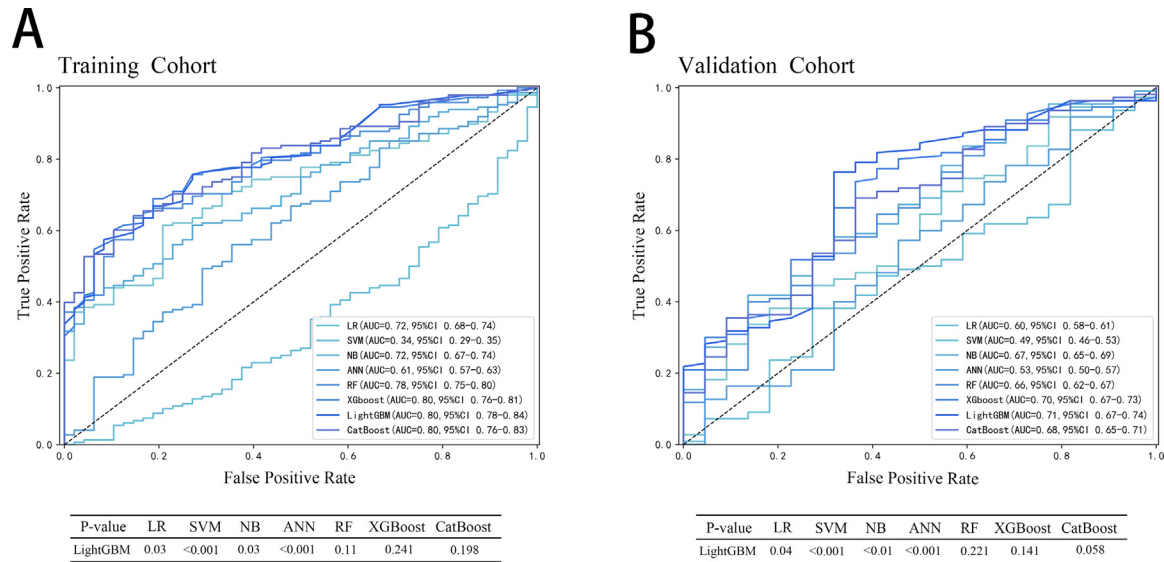


Fig 2. Receiver operating characteristic (ROC) curve of the LGBM model in the derivation and validation cohorts. **(A)** ROC curve of the LGBM model for the derivation cohort. The area under the curve (AUC) was 0.809 (95% CI 0.788-0.826). **(B)** ROC curve of the LGBM model for the validation cohort. The AUC was 0.714 (95% CI 0.698-0.725).

cohorts (Fig 3). The optimal performance was observed at a threshold of 0.3 in the training cohort and 0.25 in the validation cohort, defining a clear range in which the model provides meaningful clinical value.

This advantage was further supported by the decision curve analysis (Fig 4), where the curve remained above the “treat-none” line throughout all thresholds, indicating the absence of net harm. Together, these results highlight that the LGBM score can serve as a reliable tool to guide clinical interventions across a broad, clinically relevant range of risk probabilities.

Adjustment of eGFR Criteria for the KT Recipients

In current clinical practice, an eGFR value of 60 is commonly used as the threshold for diagnosing renal dysfunction. However, our follow-up study revealed that in KT recipients, renal function remains stable even when eGFR falls below 60. Abnormal renal function is only observed once eGFR drops below a certain lower threshold. This finding prompted our study, which aims to refine the eGFR cutoff specifically for KT patients, improving the accuracy of renal dysfunction classification within this population. The goal of our research is to

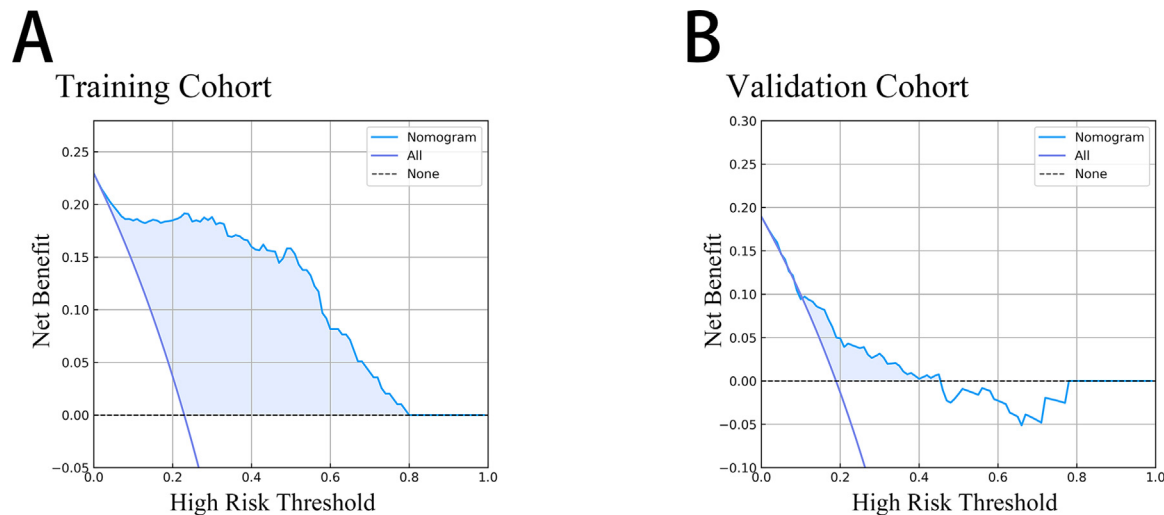


Fig 3. Decision curve analysis of the LGBM model in training and validation cohorts. **(A)** Decision curve analysis of the LGBM model for the training cohort. **(B)** Decision curve analysis of the LGBM for the validation cohort. Deep blue line: assuming no patient died within 1 year. Black dashed line: assuming all patients died within 1 year. The two lines serve as references.

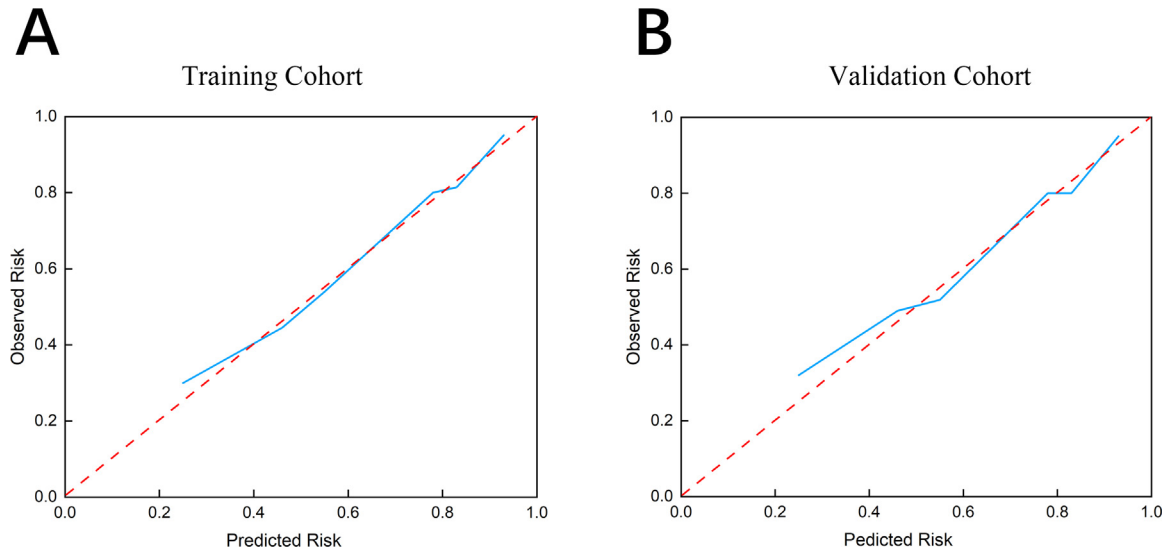


Fig 4. Calibration curve of the LGBM model for diagnosing to dangerous kidney transplantation patients. **(A)** Training cohort, **(B)** validation cohort. LGBM, light gradient boosting model.

establish a more appropriate evaluation standard for KT recipients, addressing the limitations of previous studies that failed to consider the unique characteristics of this patient group. To achieve this, we systematically assessed the predictive accuracy of various eGFR thresholds ranging from 20 to 120 for renal dysfunction in KT recipients. Through iterative adjustment of the eGFR value, we identified eGFR = 45 as the optimal cutoff.

Based on the results of the follow-up survey, KT patients were categorized into three groups: complete renal function recovery, partial renal function recovery, and renal dysfunction, with the latter being the primary focus of our study. As depicted in Fig 5, using the conventional eGFR threshold of 60 for diagnosing renal dysfunction led to a higher proportion of misclassified patients. In contrast, adjusting the threshold to eGFR = 45 significantly improved classification accuracy. Notably, eGFR values among the three groups exhibited clear distinctions. Our findings indicate that within the first 3 months following kidney transplantation, eGFR values in KT patients undergo substantial fluctuations, after which they tend to stabilize. However, patients with renal dysfunction showed a loss of eGFR data after this period.

DISCUSSION

This study aimed to develop a model capable of predicting 12-month post-transplant graft survival while exploring the relationships between graft survival, preoperative information, and postoperative indicators. Additionally, we investigated an appropriate threshold for evaluating renal dysfunction in KT patients, identifying an eGFR threshold of 45 as the optimal classification standard.

In this study, we first collected clinical data from KT patients who met the inclusion criteria and subsequently applied advanced machine learning models for modeling. Through comparative experiments conducted on the same validation set,

the LGBM model demonstrated the best performance, with an AUC value of 0.71, outperforming other models under the same conditions. The variables in the clinical data often exhibit nonlinear interactions and contain missing values. The tree structure of the LGBM model effectively captures these nonlinear relationships without requiring extensive preprocessing. The Gradient-based One-Side Sampling technique further enhances the model's ability to learn from information-rich samples, while the Exclusive Feature Bundling algorithm effectively reduces feature redundancy and improves model generalization. These innovative designs enable the LGBM model to better handle the complexity and heterogeneity of medical data. Using feature selection methods, we identified several key variables significantly associated with KT survival, including gender, blood type, donor source, HDL-C, BUN, and HGB. The experimental results indicated that the LGBM model, constructed using only these few features, was able to accurately predict the KT survival of KT patients. Furthermore, we performed an interpretability analysis of the LGBM model, quantifying the contribution of each feature to the prediction outcomes. Specifically, the contribution of features was ranked from highest to lowest as follows: HGB, blood type, HDL-C, BUN, gender, and donor source. Among these, HGB and HDL-C levels significantly reflected the renal condition of KT patients, highlighting the importance of these two variables in the prediction.

We found that these six features were significantly associated with renal function in KT patients, a result consistent with several existing studies. For example, multiple studies have suggested a higher risk of graft failure in male KT recipients [29,30]. The relationship between kidney donor characteristics and post-transplant renal function has been well-documented and incorporated into prediction models [31]. However, the link between blood type and renal dysfunction in KT patients has

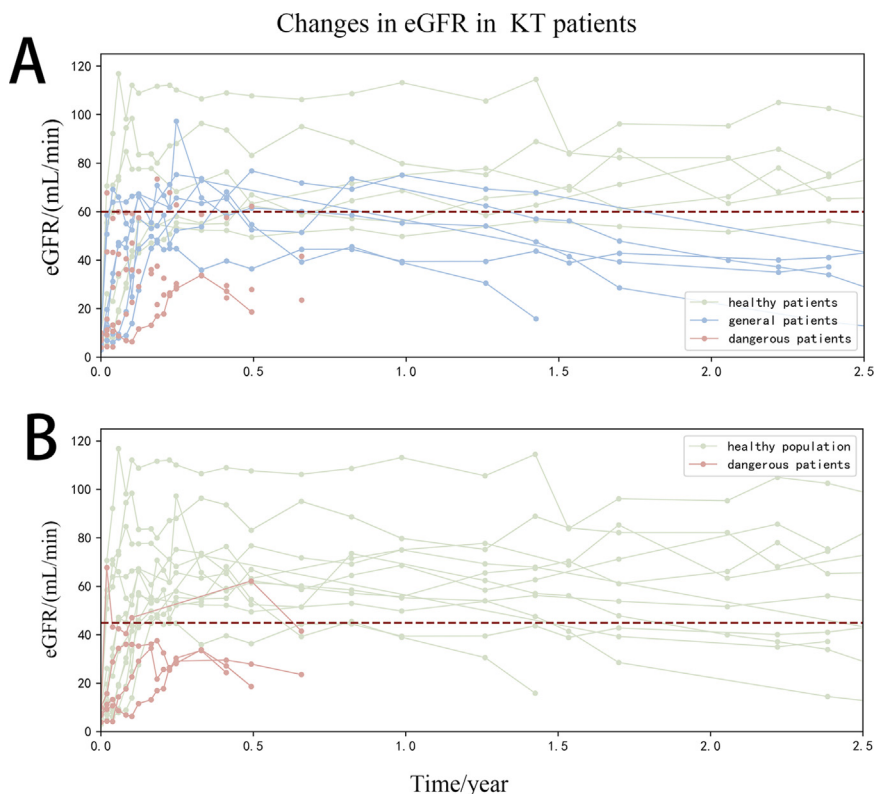


Fig 5. Changes in eGFR among KT patients with different levels of risk. **(A)** Changes in eGFR in patients with baseline eGFR = 60. **(B)** Changes in eGFR in patients with baseline eGFR = 45. It appears that classifying different patients becomes more effective when eGFR decreases from 60 to 45.

not been addressed in previous research, warranting further investigation. Dyslipidemia is commonly observed in KT patients, particularly marked alterations in HDL-C. Previous studies have identified HDL-C as a crucial predictor of post-transplant survival, noting that renal function decline may affect HDL-C levels and the efficacy of statin treatments [32–34]. The model developed in this study provides new insights into how HDL-C influences renal function in KT patients, especially through the potential modulation of HDL-C levels by pharmacological intervention, which may facilitate renal recovery. Furthermore, significant changes in BUN levels before and after kidney transplantation suggest that regulating BUN levels could have a beneficial impact on renal function recovery. HGB, closely associated with overall health, has also been identified as a valuable predictor of KT survival.

In this study, we optimized the eGFR threshold for identifying kidney dysfunction in KT patients to improve classification accuracy. For the general population, an eGFR below 60 typically indicates kidney dysfunction. However, for KT patients who have experienced CKD, the eGFR standard tends to be lower even when kidney function is normal, and there is no unified threshold for diagnosing kidney dysfunction in these patients. Our experimental results indicate that an eGFR threshold of 45 is more appropriate for KT patients, with a classification accuracy of 68%, which is 14% higher than the conventional eGFR threshold of 60 (Fig 6).

Despite providing valuable insights, this study acknowledges several limitations. First, the LGBM model was developed

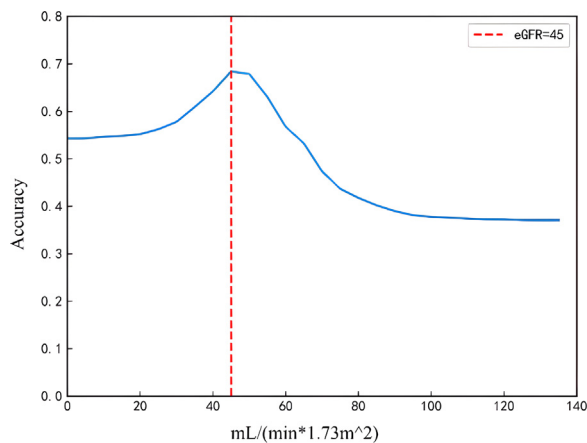


Fig 6. The accuracy curve under different eGFR criteria.

primarily using data from Chinese patients, necessitating further validation in other populations to ensure broader applicability. Second, while the model was tailored for KT patients without viral hepatitis, its performance in patients with other chronic conditions remains to be evaluated. Third, due to limited transplant data, our analysis was based on a cohort of approximately 300 high-quality datasets. Future studies should focus on expanding the dataset to enhance the model's robustness and predictive accuracy.

In summary, this study successfully developed an LGBM-based predictive model for accurately estimating 12-month graft survival in KT patients. DCA and calibration curve validations further demonstrated the effectiveness of the LGBM model in assessing the severity of renal dysfunction in KT patients. With its high discriminative power and stability, the model not only assists clinicians in optimizing patient management but also serves as a potential noninvasive tool for evaluating the severity of renal dysfunction in KT recipients.

DISCLOSURE

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DATA AVAILABILITY

The authors do not have permission to share data.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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